

Square Yards: Redefining the real estate tech landscape on a robust cloud platform

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ABOUT SQUARE YARDS

To support the exponential growth in traffic, Square Yards needed a robust cloud platform that provided the flexibility and customer support to thrive in an agile environment.



About Square Yards

Square Yards is India's largest integrated real estate marketplace, with category leadership presence across multiple touchpoints of consumer home buying and ownership journey. With urbanization and rising disposable incomes as the core theme, Square Yards, with over 8 million monthly traffic and over US\$3 billion GTV, is the largest and asset-light proxy play to the growing residential demand story of India. One of the few Indian start-ups to taste global success with a presence in 100+ cities across 9 countries, Square Yards is at the forefront of

Google Cloud results

- Improves page load performance by up to 25% with GKE and Pub/Sub
- Completes a full-stack migration with zero downtime
- Reduces cost by up to 15% with Google Cloud autoscaling
- Enables flexibility with custom VMs

Improve traffic 50%

tech adoption in the sector, with multiple patents across VR and AI domains.

Industries: Real Estate & Construction

Location: India

(<https://cloud.google.com/compute/>),

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/kubernetes-engine/>),

BigQuery (<https://cloud.google.com/bigquery/>)

(<https://cloud.google.com/pubsub/>) Google

Compute Engine

(<https://cloud.google.com/compute/>)

(<https://cloud.google.com/compute/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

(<https://cloud.google.com/products/iam/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Google Workspace

(<https://workspace.google.com/>)

Google Drive

(<https://workspace.google.com/products/drive/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

The residential real estate market in India has seen tremendous growth

(<https://www.livemint.com/money/personal-finance/future-of-real-estate-market-in-india-in-2023-11676368024008.html>)

since the pandemic, as people are starting to spend more time at home. With hybrid work becoming a

norm, people are looking for bigger rooms and extra space. According to Vivek Agarwal, Co-founder and CTO at Square Yards (<https://www.squareyards.com/>), "People want bigger homes, and homes with extra rooms as they've started to realize the value of a home more."

Supporting the uptick in demand in the residential real estate space, Square Yards has seen unprecedented growth, with close to eight million monthly visitors on its platform. This meant that it needed to improve its current infrastructure to support a higher influx of users while ensuring it maintains reliability for a great customer experience.

The company turned to Google Cloud (<https://cloud.google.com/>) for its reliability and scalability, as well as its flexible infrastructure. More importantly, Agarwal says that working with a cloud platform that would be able to provide the customer support they needed was just as important, something that was lacking in the organization's previous cloud provider.

Going beyond technicalities for a seamless user experience

As a cloud-native platform, Square Yards is familiar with operating in the cloud. However, the company needed more than just a cloud provider. It needed support from people who were knowledgeable about the industry and were able to provide the right advice on using the most appropriate technologies according to the circumstances.

"Whenever we were facing any challenges, we had to go through many layers of hierarchy with our previous cloud provider to get to the appropriate person and get our queries resolved," explains Agarwal. "Instead of just reaching out to an engineer for support, we had to file a ticket on the system. When it gets picked up, we would have to explain the entire situation to them again, which was frustrating."

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*—Deepak Kushwaha,
Principal Partner and*

*Head of Engineering,
Square Yards*

With Google Cloud, Agarwal says that this is no longer an issue as they are always able to connect with someone on the team who would provide the advice needed for the business. "Even if we have a question about a product that is not a Google Cloud product, the Google Cloud team is able to help us by providing insight on the best way to use it so we have a better understanding."

Improving website reliability and availability to grow user base

Since migrating to Google Cloud, Square Yards has observed a 50 percent improvement in its web traffic. "Optimizing our web traffic was a key criteria for us when moving to Google Cloud because we gain customer acquisition from users who visit our website," explains Agarwal.

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vertical scaling.
We found
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Engine to be
very easy to use,
and the
performance
has been
exceptionally
good."

—Vivek Agarwal, Co-
founder and CTO, Square
Yards

After moving to Google Cloud, Square Yards has been able to improve its page load performance by 25 percent. "Google Cloud gives us the flexibility to create our custom VMs on [Compute Engine](https://cloud.google.com/compute) (<https://cloud.google.com/compute>) and decide how much RAM or CPU we want to add," says Vivek Agarwal, Co-founder and CTO, Square Yards. "As the number of users grows over time alongside web traffic, we need to do more horizontal scaling, alongside the vertical scaling. We found [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine) (<https://cloud.google.com/kubernetes-engine>) (GKE) to be very easy to use, and the performance has been exceptionally good."

To date, Square Yards has converted 12 of its monolithic services into a microservices architecture and deployed them on GKE clusters. "With GKE able to function on autopilot, we are able to get the best out of these clusters, optimizing the performance, speed, and overall website traffic which has had a direct impact on our bottom line," says Agarwal. The team also leverages [Pub/Sub](https://cloud.google.com/pubsub) (<https://cloud.google.com/pubsub>) to integrate multiple data pipelines for a smoother user experience and runs [Backup and DR Service](https://cloud.google.com/backup-disaster-recovery) (<https://cloud.google.com/backup-disaster-recovery>) to protect its workloads.

With the autoscaling capabilities of Google Cloud, Square Yards has been able to reduce its operational cost by 15 percent. "This percentage doesn't even include the promotional discounts that were available to us," says Agarwal.

Completing a full-stack migration with zero downtime

Kushwaha describes the migration from the previous cloud provider to Google Cloud as one that was "no less than performing heart surgery." With users across India, the UAE, the Middle East, Australia, and Canada, customers are accessing the Square Yards platform 24/7 due to the time difference, so being always available was crucial.

Together with Google Cloud, the engineering team at Square Yards divided the migration into three phases. They set up everything in parallel, so all they needed time for was the DNS propagation.

"We had expected this to be a long and laborious journey with hiccups in between, but the overall project management was smooth beyond expectations. The best part about the migration was that the experience was so seamless for users, they never would have known that we had completed a migration to a different cloud platform," says Kushwaha.



Square Yards' team retreat

Shaping the future of real estate with AI and ML

"We're excited about using AI to transform the real estate industry, and with the strong support of Google Cloud, I believe there is a lot more we can do together to provide the best experience for our users," says Hitesh Singla, Co-founder and CIO at Square Yards.

Square Yards is well on its way to shaping the real estate industry by developing immersive digital twins of key cities. "In the future, we want people to be able to fully immerse themselves in a city when searching for property. So instead of a 2D experience based on a map, they would be able to feel like they're walking down the actual street, looking at the properties as if it were real life," explains Singla.

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The team already has plans to launch this new digital experience in Dubai and Toronto and will continue to work with Google Cloud to develop the most seamless experience for every user.

Square Yards also recognizes that data is key to the success of the company, and uses BigQuery (<https://cloud.google.com/bigquery>) to ingest and

capture data from every user interaction on its platform for lead segmentation and lead scoring.

"We capture the interactions happening on our website and based on this data, we are able to understand the different demographics and their needs better," Singla shares.

Making the work experience seamless across platforms with Google Workspace

As the company grows, so does the number of employees. Already using [Google Workspace](https://workspace.google.com/) (https://workspace.google.com/) for collaboration, Square Yards moved from Business Standard to the Enterprise plan to ensure they get the best features available. Employees use [Google Drive](https://workspace.google.com/products/drive/) (https://workspace.google.com/products/drive/) daily as a single source of truth to collaborate and share documents.

Security was a key factor that drove Singla and his team to implement Google Workspace Enterprise, as they work with a lot of sensitive data for customers.

"The storage and security that Google Workspace offers with Vault, DLP, data regions and enterprise endpoint management were big drivers for us because, with Google Workspace, we no longer need to deploy third-party security applications," says Singla.

Beyond data storage and security, all meetings at Square Yards are conducted in [Google Meet](https://workspace.google.com/products/meet/) (https://workspace.google.com/products/meet/) , which has been a game changer for the company since

they have teams across nine countries and more than 50 offices. "With Meet, we don't have to worry about where people are physically, as all they need to do is tune in from their devices," says Singla, adding that the bandwidth friendliness of the platform helps with the seamless connectivity.